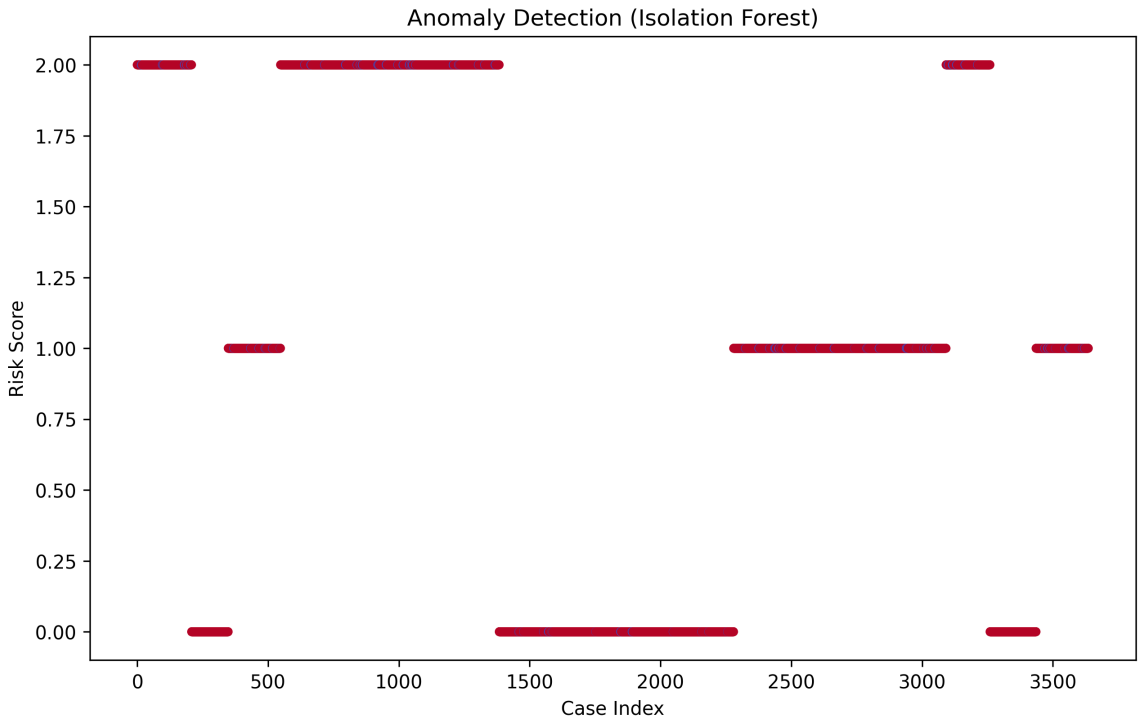
# PulmoBench Comprehensive Analytical Intelligence Report

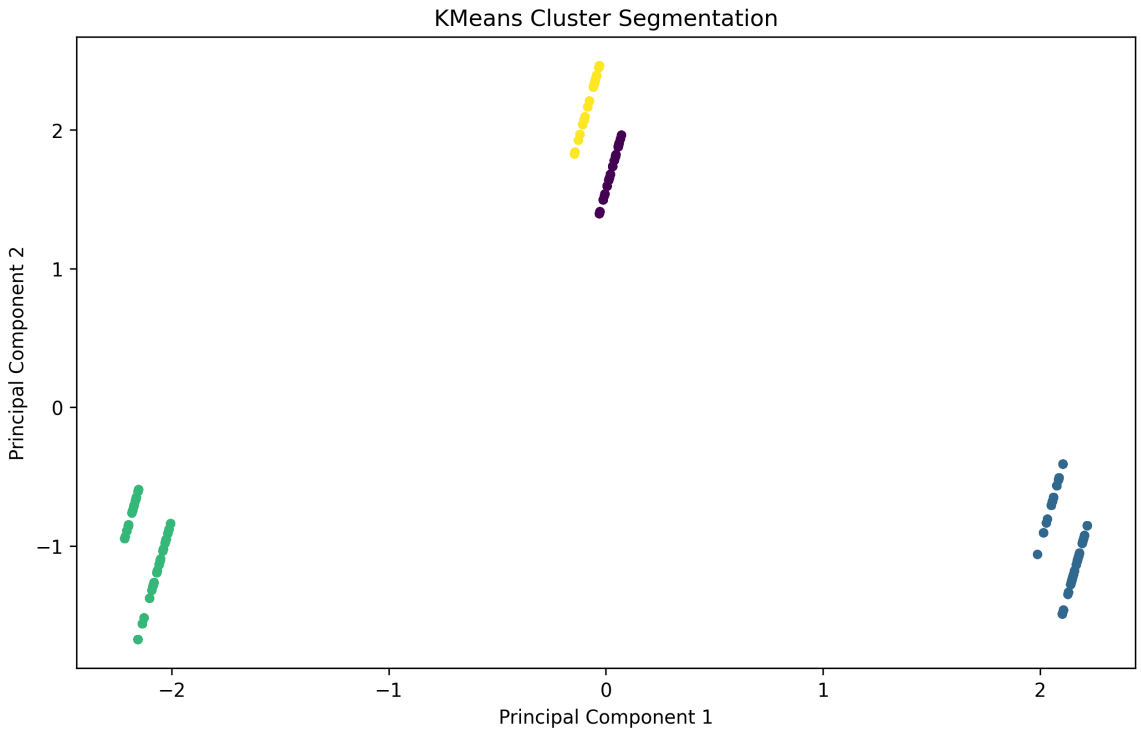
This report analyzes 3636 patient records using dimensional reduction, clustering, anomaly detection, spectral analysis, and statistical profiling. The silhouette score of 0.326 indicates moderate structural separation between identified clusters.

## Anomaly



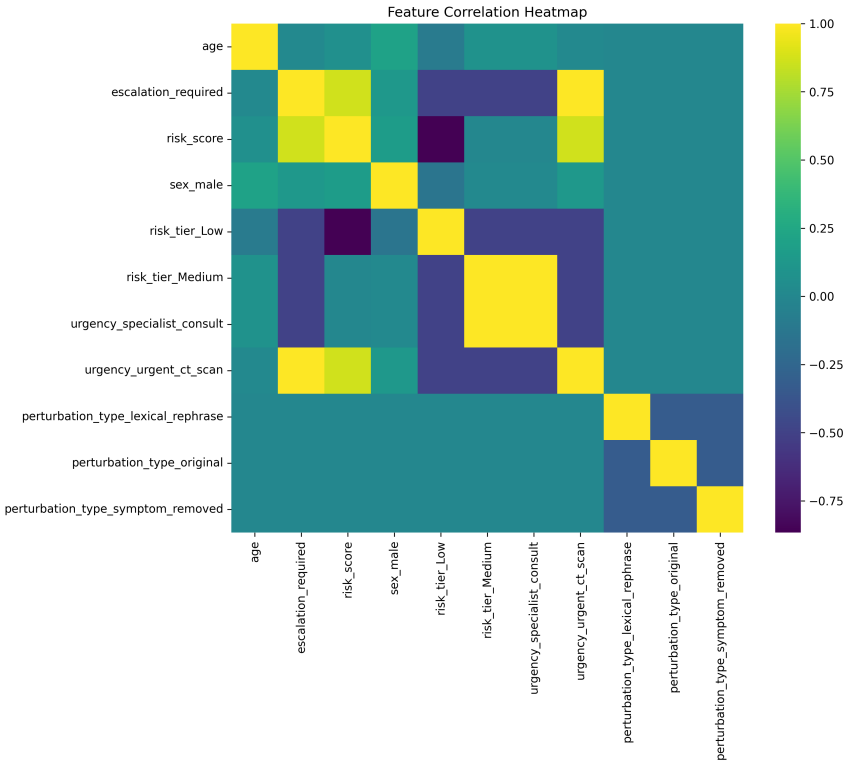

Highlights statistically rare cases based on multivariate isolation depth.

## Cluster



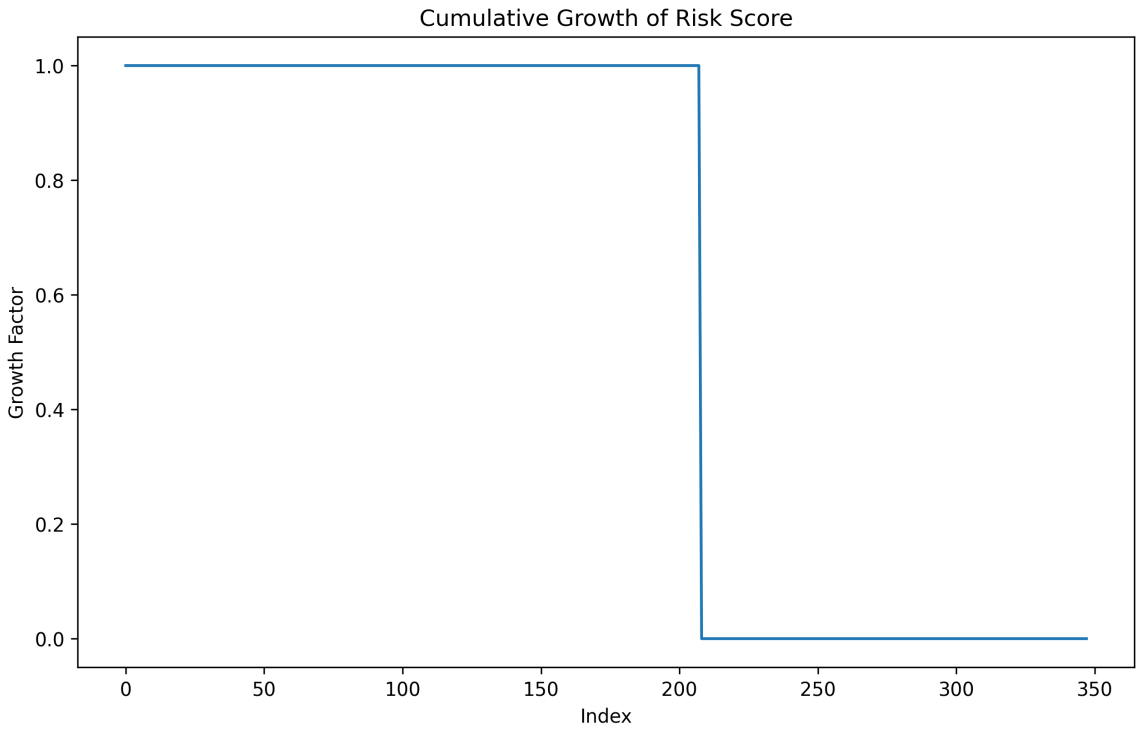

Segments population into 4 archetypes. Silhouette score 0.326 measures cluster cohesion.

# Correlation



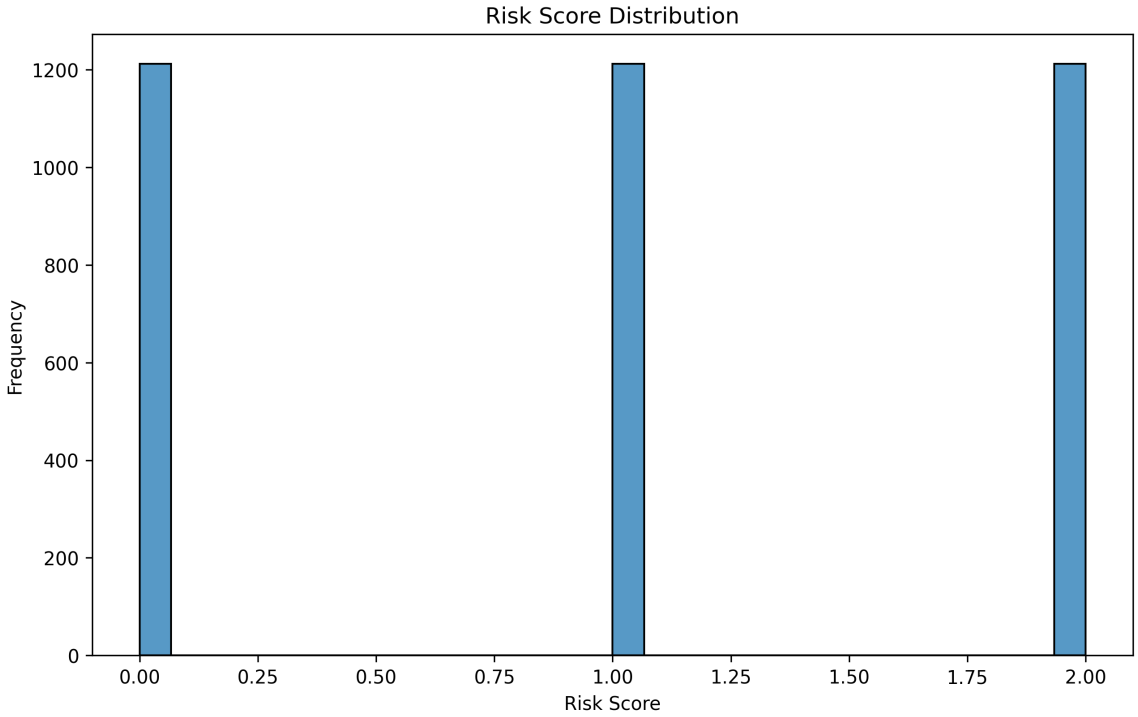

Quantifies linear interdependencies among predictors.

## Cumulative Growth



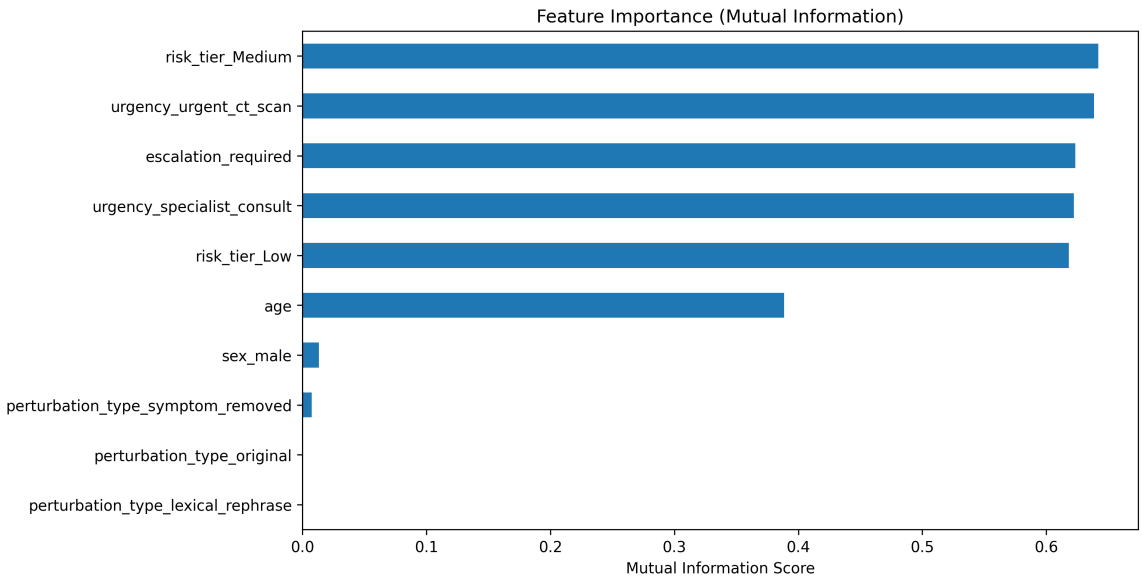

Shows compounded proportional change in risk trajectory.

## Distribution



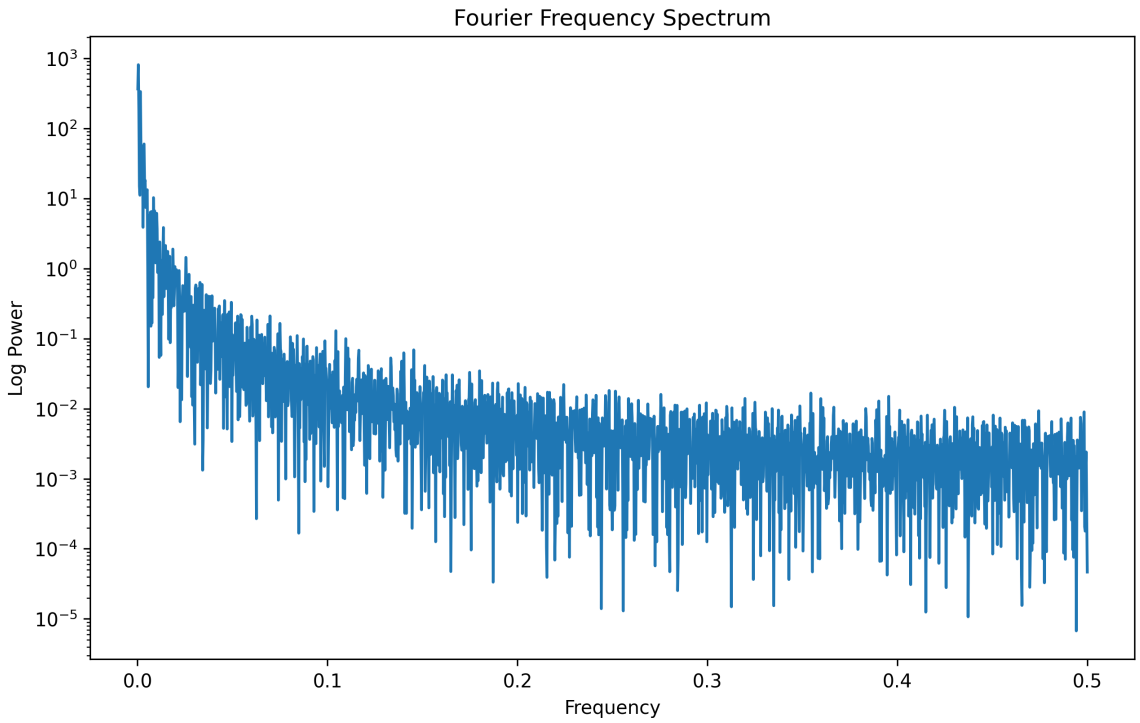

Displays central tendency and skewness of risk score.

# Feature Importance



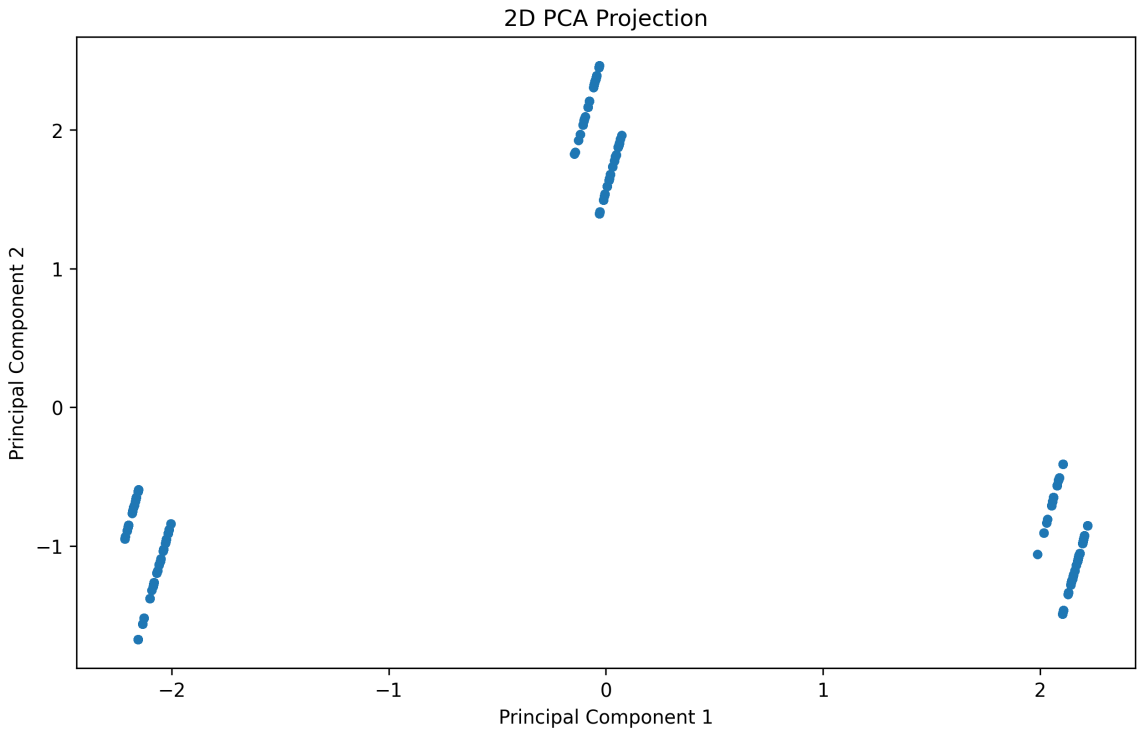

Ranks nonlinear predictive influence of features on risk score.

## Fourier



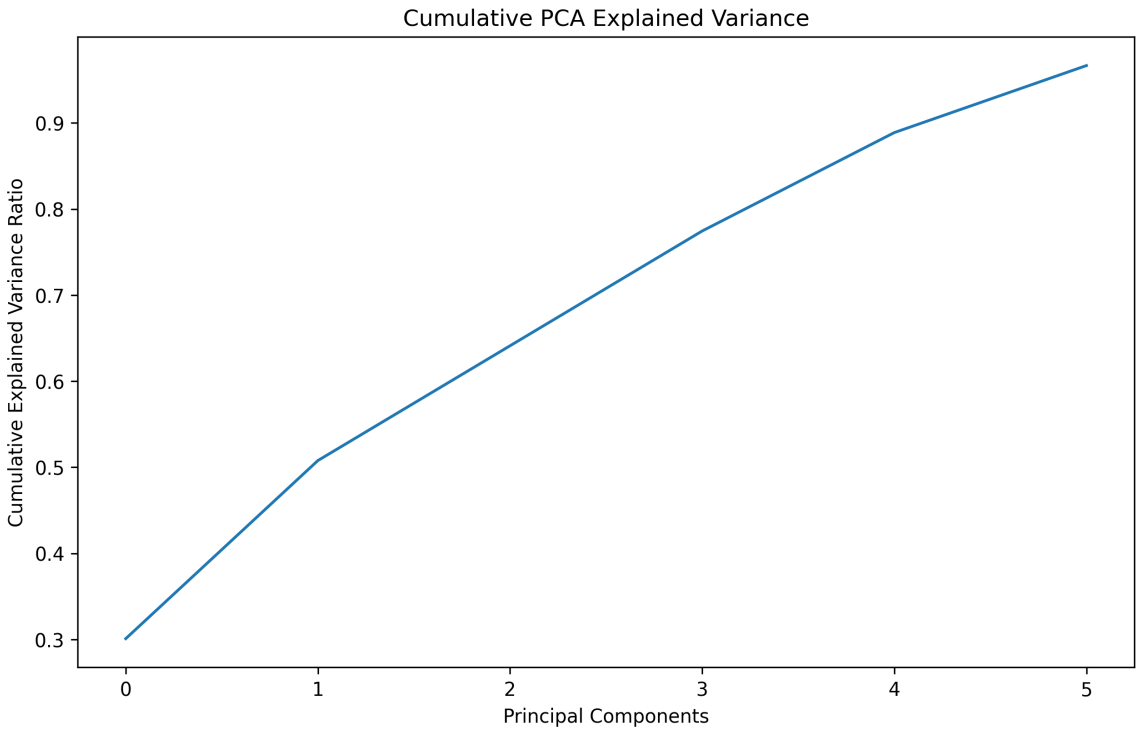

Reveals periodic structure embedded in risk score ordering.

## Pca Projection



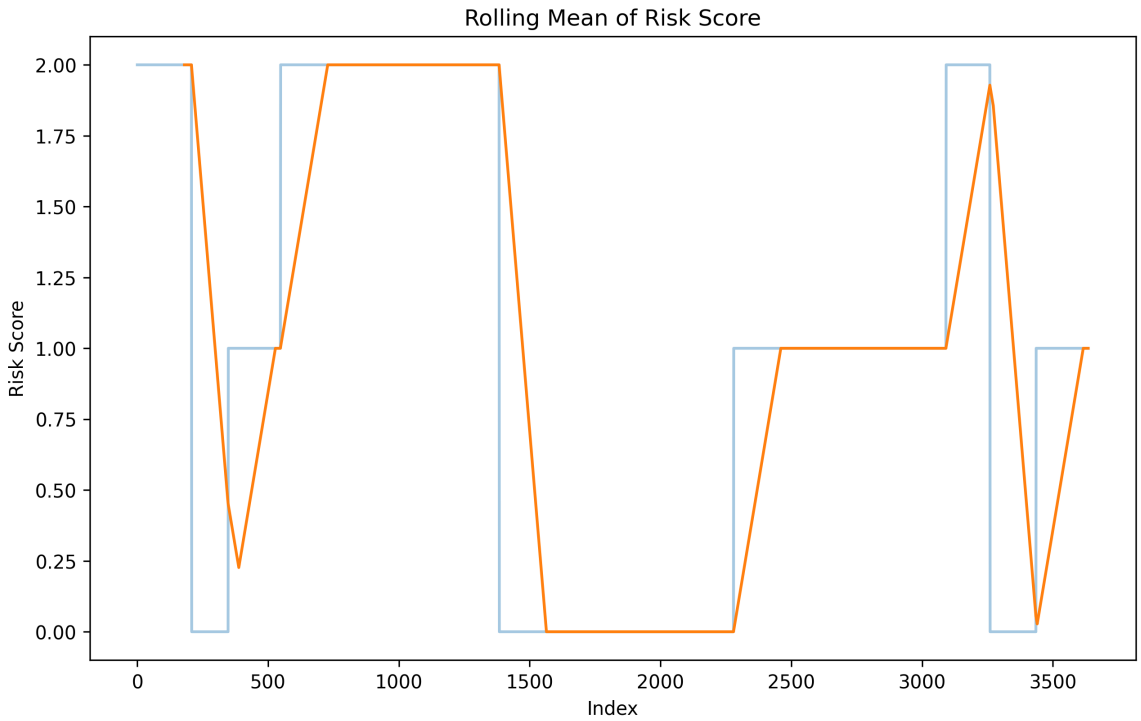

Visualizes structural grouping in reduced dimensional space. Cluster separation indicates population heterogeneity.

## Pca Variance



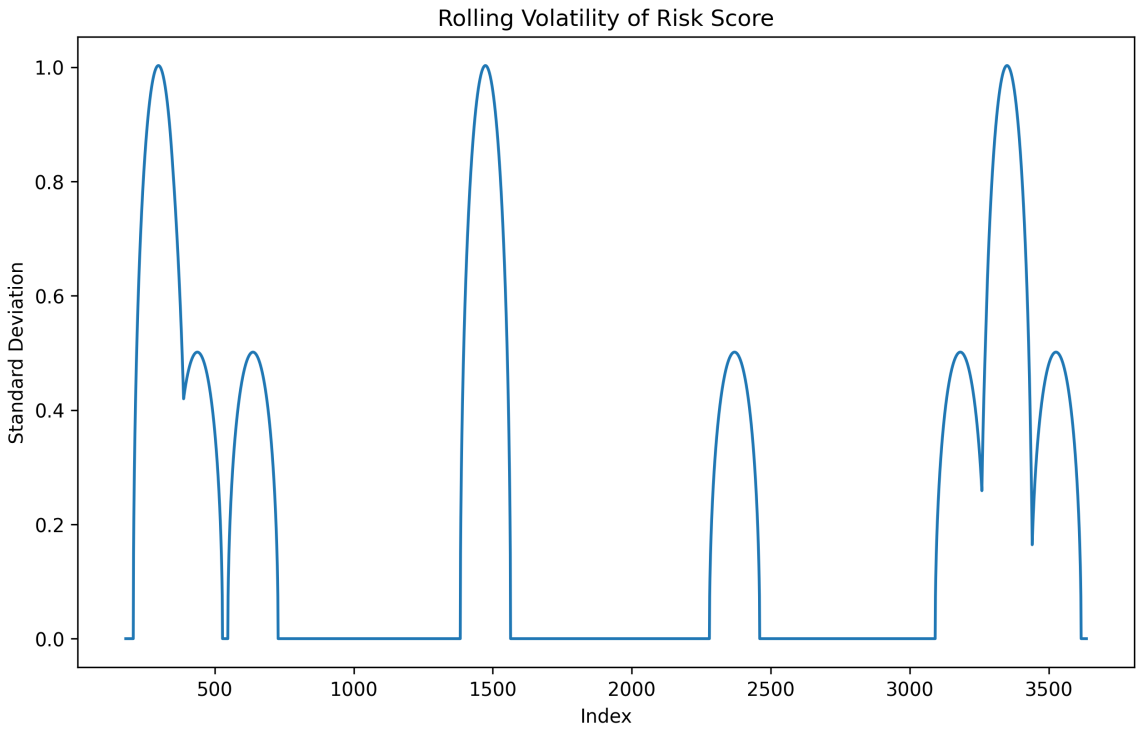

Explains how variance accumulates across principal components. Early saturation indicates strong latent dimensional structure.

## Rolling Mean




Smooths short-term fluctuations to expose longer-term trend patterns.

## Rolling Volatility




Measures temporal instability in risk behavior.